

Answer 蒙到你了? 推断一些元素; 积累一点常识; 练习几个积分; /
学习一些晶体.

I. Monazite

每空一分

1-1-1: A: $\text{La}(\text{NO}_3)_3 \cdot 2(\text{NH}_4)\text{NO}_3 \cdot 4\text{H}_2\text{O}$

B: ~~Ce~~ Ce

C: $\text{Ce}(\text{OH})_4$

D: CeO_2

E: Pr

F: Pr_2O_3

G: Sm

1-1-2: 1:1:Eu

1:Y

J: Tb

K: Ho

L: Tb_4O_7

1-1-3: $\text{Sc}(\text{C}(\text{CF}_3\text{COO})_3)$ 3分

1-2: 分析略, 为 2分

$\text{Na}[\text{Yb}(\text{C}_2\text{O}_4)_2] \cdot \text{H}_2\text{O}$ 结果2分

$$n(\text{Ce}^{4+}) = n(\text{Ce}) = \frac{247.3546}{F} = 2.56364 \times 10^{-3} \text{ mol}$$

$$n(\text{KMnO}_4) = cV = 2.05 \times 10^{-3} \text{ mol}$$

$$\text{对左 } 8.204 \times 10^{-4} \text{ mol 萃取}$$

$$M_1: \text{C}_2\text{O}_4 = 1:2, \text{可以打出 } \text{Na}[\text{Yb}(\text{C}_2\text{O}_4)_2] \cdot \text{H}_2\text{O}$$

1-3: 设选过了 dV 体积的煤油, 水相为 L , 总量为 $n(V)$

$$\text{萃取量: } dV \cdot c(\text{有机相}) = dV \cdot \frac{n(V)}{L} \cdot \beta = -dn \rightarrow \text{水相减少的量,}$$

$$\frac{n(V)\beta}{L} = -\frac{dn}{dV} = n\beta \quad \frac{dn}{n} = -\beta dV \quad \ln\left(\frac{n_1}{n_0}\right) = -\beta V$$

$$\text{代入 } \frac{n_1}{n_0} = 1 - 99.99\% = 10^{-5}, \beta = 10^{3.1} \text{ 解得 } V = 9.145 \times 10^3 \text{ L, 可见萃取十分高效.}$$

(5')

II. Laser:

2-1 解析兄弟一张答案

2-2-1: Y(8), Al(4.6), O(4); 0.4种 (一点2分) Al(4.6):6(2) = 3:2; O(4):4(2) = 1:1

2-2-2: 4.55 g cm^{-3} (3); 因为其晶格紧凑, 机械强度与导热性能优异 (2')

2-3-1: 让 $I(n)$ 为 n 次往返后光强, $I_1 = I(0)$ 能不出光, 则 $I_{\text{出}} = I_{\text{衰减}} = I_0$

$$I(1) = 0.95 \times 0.99 I_0$$

$$I(n) = (0.95 \times 0.99)^n I_0$$

$$I_{\text{出}}(n) = 0.05 I(n)$$

$$I_{\text{出}} = \sum_{n=0}^{\infty} I_{\text{出}}(n) = 16.806 \times 0.05 I_1 = 0.8403 I_1$$

(6')

$$I_1 = 1.19 I_0, I_{\text{出}} = \sum I(n) = 20 I_0$$

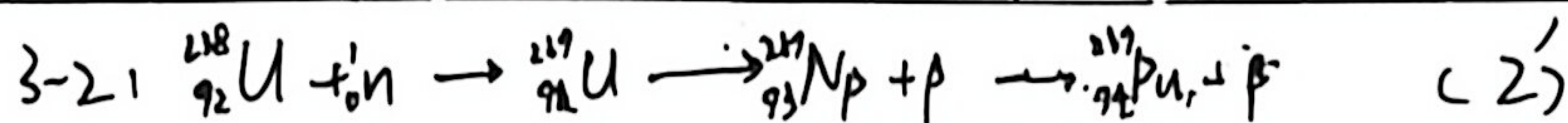
$$\frac{I_{\text{出}}}{I_{\text{入}}} = \frac{8\pi h\nu^3}{c^2} \cdot \frac{1}{20 I_0} = \frac{8\pi h\nu^3}{20 I_0 c^2}$$

可见有强光的时候自发辐射占比很少,

2-3-2: 如图可见晶体明显由 BO_3^{3-} 层与层间 K^+ 组成。实际上, 该晶体早年即因其在紫外光发射为人所知, 然而, 其层状生长倾向太严重, 导致有效进展在21世纪由我国科学家领头研究时才取得,

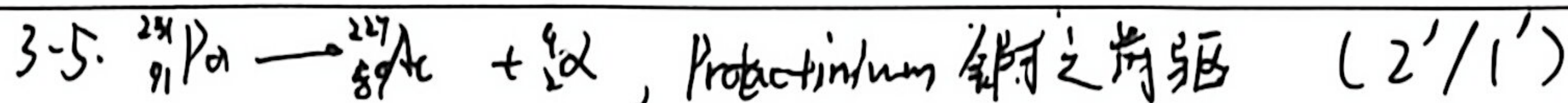
(层状结构 3')

III: 3-1. Pu-238 / Pu-239 (2')



3-3. 核中核子中子的成对 (2')

3-4. 衰变产物 (1')



3-6. 巨大的对光子(电磁作用)散射强, 高能电子经过巨大核会产生韧致辐射

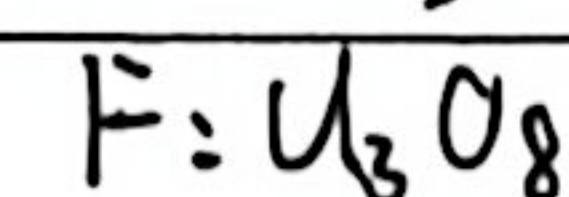
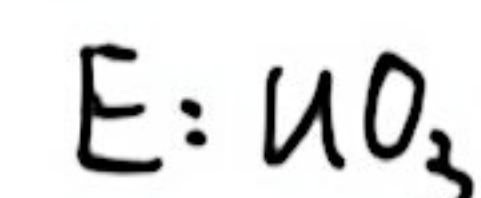
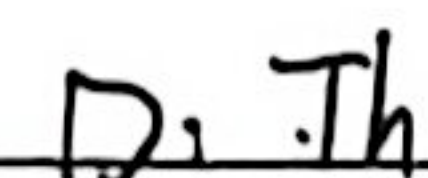
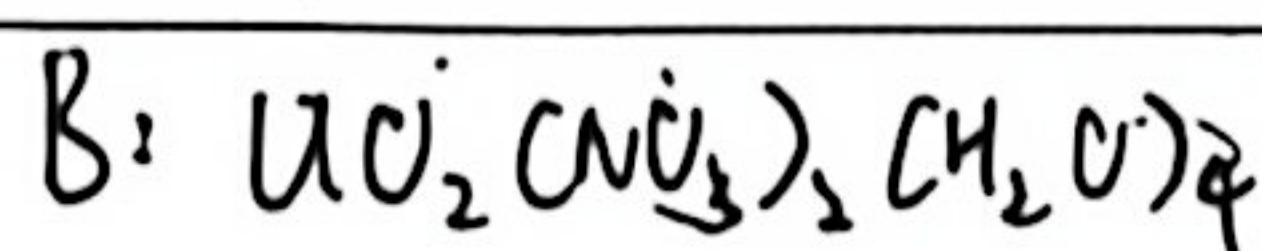
(Bremsstrahlung); 中子反射材料原子核大, 密度高, 中子撞击频率大, 功能损失小。

但倘若要吸收中子, 请使用含氢量高的物质, 因为氢核与中子质量接近,

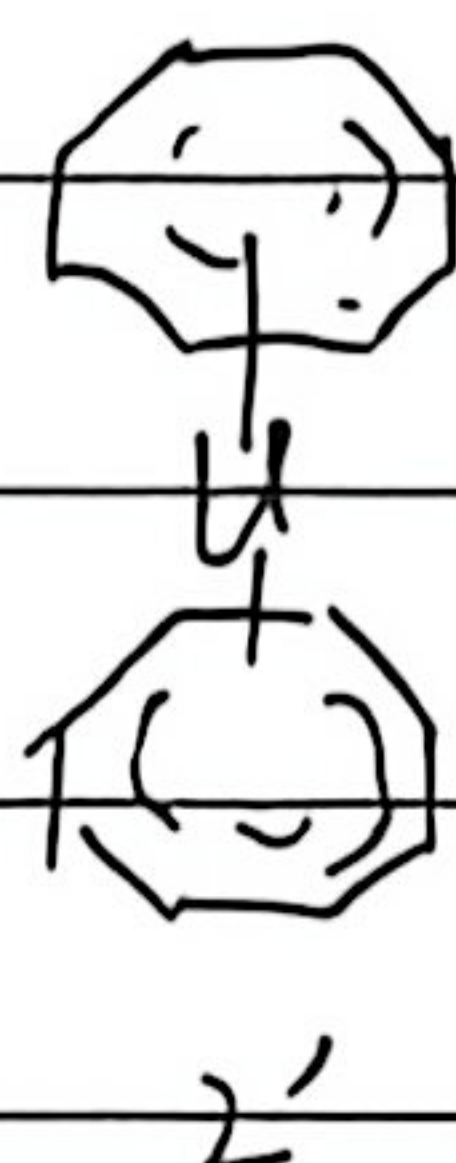
弹性碰撞时动能传递多 (4')

每克分

IV: 4-1: A, U

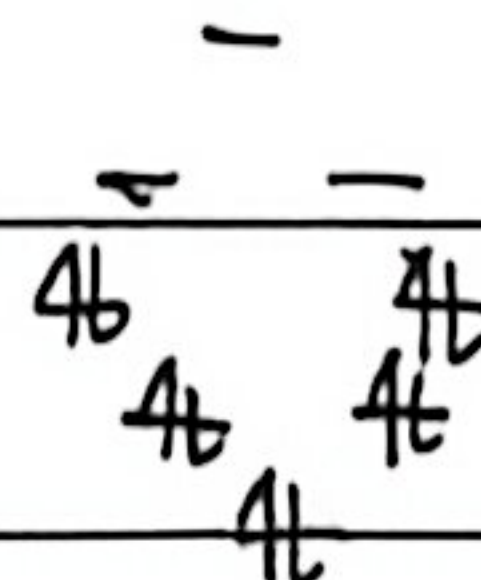


4-2-1



分子结构示意图 4'

示出:

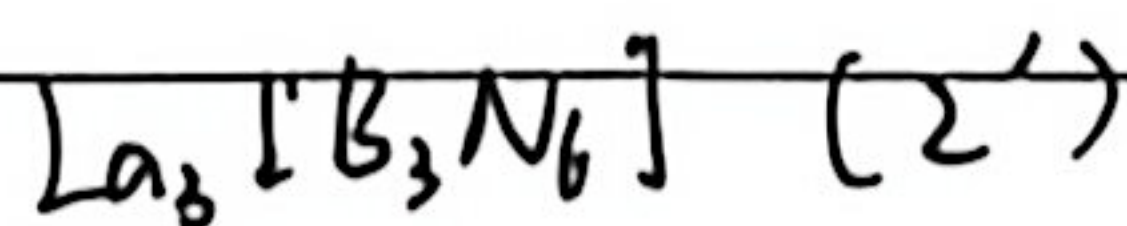
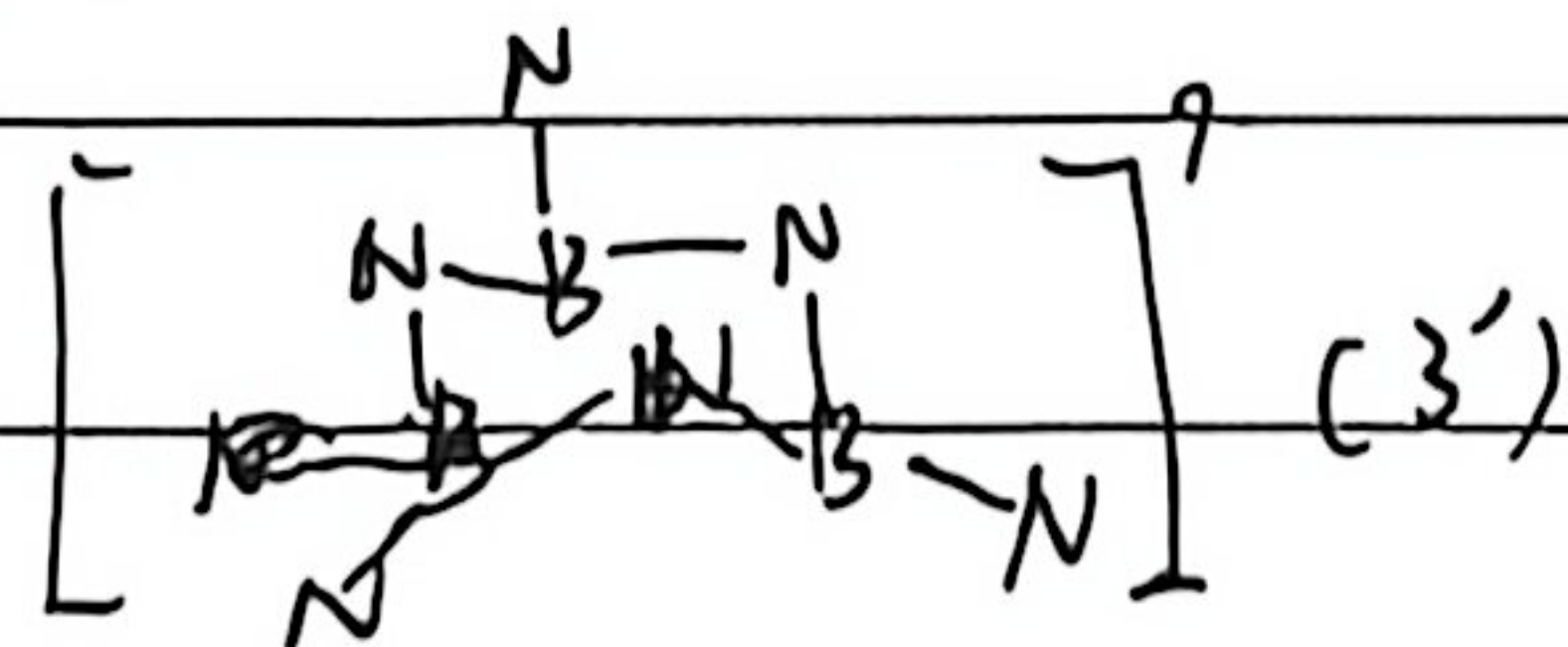


作用与



作用即可。

4-2-2:



4-2-3: ①亲水性: 亲水侧链(PEG)与叶林类似的叶林

(2') 环

②较短半衰期: ①亲水侧链在血液与肾脏中多, 排出快

(2')

②叶林环易被人体代谢

因为 La^{3+} 大小适中, La^{3+} 结合力太强而 Lu^{3+} 太强

(2')

声明: 用CASIO积分者不扣分, 手折积分者一个积分加10分

2-1 答案. 会一点量子力学还是较为重要, 尤其是对一些简单的体系.

2-1-1: $\psi(x, t) = \varphi(x) \psi(t)$

$$\psi \cdot i\hbar \frac{\partial \psi}{\partial t} = \varphi \hat{H} \varphi$$

$$\frac{1}{\varphi} i\hbar \frac{\partial \psi}{\partial t} = \frac{1}{\varphi} \hat{H} \varphi = E, \text{ 这是两个分离的方程.}$$

① $i\hbar \frac{\partial \psi}{\partial t} = E\psi$ $\varphi = \exp(-\frac{Et}{\hbar})$ ② $\hat{H}\varphi = E\varphi$ 定义 Schrödinger 方程

$$\frac{d\psi}{dt} = \frac{E}{i\hbar} \psi$$

$$\frac{d\psi}{\psi} = \frac{E}{i\hbar} dt$$

$$\ln \psi = \frac{Et}{i\hbar} \text{ (不考虑初始条件)}$$

(4', 一个方程 2')

2-1-2: $\psi = \sum a_n(t) \psi_{n0}$

LHS: $i\hbar \frac{\partial}{\partial t} \sum a_n(t) \psi_{n0} = i\hbar \sum \dot{a}_n(t) \psi_{n0} + i\hbar \sum a_n(t) \frac{\partial \psi_{n0}}{\partial t}$

这是原体系的解.

RHS: $\hat{H}_0 \sum a_n(t) \psi_{n0} + \hat{H}_p(t) \sum a_n(t) \psi_{n0} = \sum a_n(t) \hat{H}_0 \psi_{n0} + \sum a_n(t) \hat{H}_p(t) \psi_{n0}$

$$i\hbar \sum \dot{a}_n(t) \psi_{n0} = \sum a_n(t) \hat{H}_p(t) \psi_{n0}, \text{ 应用正交性 } \langle \psi_i | \psi_j \rangle = \delta_{ij}$$

$$i\hbar \sum \dot{a}_n(t) \int \psi_{n0}^* \psi_{n0} dr = \sum a_n(t) \int \psi_{n0}^* \hat{H}_p \psi_{n0} dr \quad \text{注: } \dot{a}_n(t) = \frac{da_n(t)}{dt}$$

$$i\hbar \dot{a}_n(t) = \sum a_n(t) \int \psi_{n0}^* \hat{H}_p \psi_{n0} dr$$

(6')

$$\therefore \dot{a}_n(t) = \frac{1}{i\hbar} \sum a_n(t) \int \psi_{n0}^* \hat{H}_p \psi_{n0} dr$$

2-1-3: 应用简化条件, 得:

$$\dot{a}_m(t) = \frac{1}{i\hbar} \int \psi_{m0}^* \hat{H}_p \psi_{n0} dx \quad \hat{H}_p = E_0 \cos \omega t \cdot ex$$

我们期望分离时间相关项

$$\frac{da_m(t)}{dt} = \frac{1}{i\hbar} \varphi_{m0}^* \varphi_{n0} \int \psi_{m0} E_0 \cos \omega t ex \psi_{n0} dx$$

$$= \frac{1}{i\hbar} \varphi_{m0}^* \varphi_{n0} E_0 \cos \omega t e^{i\frac{E_m - E_n}{\hbar} t} \left[\int \psi_{m0} x \psi_{n0} dx \right], \text{ 用 } \langle x \rangle \text{ 简化}$$

$$da_m(t) = \frac{1}{i\hbar} \exp(-\frac{E_m t}{\hbar}) \exp(\frac{E_n t}{\hbar}) E_0 \cos \omega t \cdot e \cdot \langle x \rangle \cdot dt$$

两次积分, 得

(8')

$$a_m = \frac{E_0 e \langle x \rangle}{2} \left\{ \frac{\exp(\frac{E_m - E_n - \hbar \omega}{\hbar} t)}{E_m - E_n - \hbar \omega} + \frac{\exp(\frac{E_m - E_n + \hbar \omega}{\hbar} t)}{E_m - E_n + \hbar \omega} \right\} + C$$

当 $t=0$ 时, a_m 也为 0., 故 (3) 可知,

$$a_m = \frac{E_0 e \langle x \rangle}{2} \left\{ \frac{1 - \exp\left[\frac{(E_m - \hbar\omega)t}{i\hbar}\right]}{E_m - \hbar\omega} + \frac{1 - \exp\left[\frac{(E_m + \hbar\omega)t}{i\hbar}\right]}{E_m + \hbar\omega} \right\}, \quad P = a_m^2$$

可以这样

2-1-4: 当 $\hbar\omega = E_m - E_n$ 或 $\hbar\omega = E_n - E_m$ (假设 $m > n$), P 都会趋近于正无穷. 虽然这引发了奇点, 但是可见频率匹配时吸收或受激辐射最强, 这也是激光的核心要求, 线宽尽量窄. (3')

2-1-5: 注意! 一维势箱乃初变要求, 必须会算! 我们主要算 $\langle x \rangle$ 项

$$\psi_n = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi}{L}x\right) \rightarrow \psi_m = \sqrt{\frac{2}{L}} \sin\left(\frac{m\pi}{L}x\right)$$

$$\langle x \rangle = \int_0^L \psi_n \psi_m x \cdot dx = \frac{2}{L} \int_0^L \sin\left(\frac{n\pi}{L}x\right) \sin\left(\frac{m\pi}{L}x\right) \cdot x \cdot dx$$

积分过程较复杂, 可见另一张原稿.

$$\text{得 } \langle x \rangle = \left\{ \frac{x[e^{i\omega_1 x} - e^{i\omega_2 x}]}{i\omega_1} + \frac{e^{i\omega_1 x} + e^{i\omega_2 x}}{\omega_1^2} - \frac{e^{i\omega_2 x} + e^{i\omega_1 x}}{\omega_2^2} - \frac{x[e^{i\omega_2 x} - e^{i\omega_1 x}]}{i\omega_2} \right\} \Big|_0^L$$

$$(\omega_1 = \frac{(m+n)\pi}{L}, \omega_2 = \frac{(m-n)\pi}{L}, m \neq n)$$

(8')

= 0, 可见其理没有先吸收又发射.

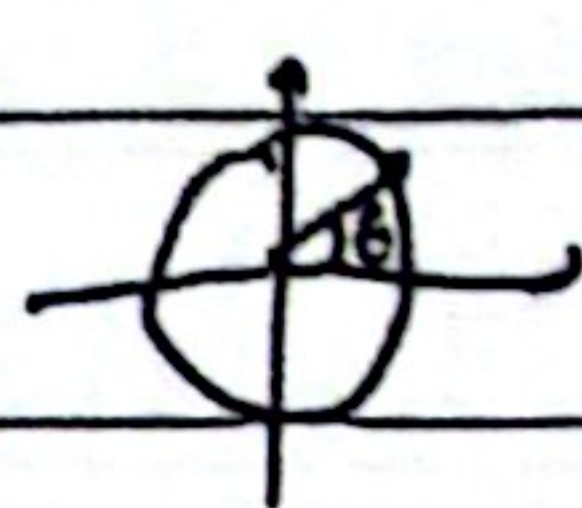
2-1-6: 用 $\hat{H} = \frac{p^2}{2m}$ 代入 $E\psi = \hat{H}\psi$, 得

$$E\psi = -\frac{\hbar^2}{2m} \cdot \frac{1}{R^2} \frac{\partial^2 \psi}{\partial \theta^2}, \text{ 代入通解 } A\sin\omega\theta + B\cos\omega\theta,$$

$\omega = \frac{r}{\hbar} \sqrt{2mE}$, 倘若 $\psi(0) = \psi(2\pi)$ (循环边界), ω 为一系列整数 $n \in \mathbb{N}$.

$$E = \frac{\hbar^2 n^2}{2mR^2}, \text{ 归一化: } 1 = \int_0^{2\pi} \psi_n^2 \cdot d\theta = B^2 \int_0^{2\pi} \cos^2(n\theta) \cdot d\theta, B = \frac{1}{\sqrt{\pi}}$$

$$\begin{cases} \psi_n(\theta) = \frac{1}{\sqrt{\pi}} \cos(n\theta) \\ E = \frac{\hbar^2 n^2}{2mR^2} \end{cases}$$



$$x = R \cos \theta, \quad \theta$$

(10')

$$\langle x \rangle = \int_0^{2\pi} \cos(n\theta) R \cos \theta \cos(n\theta) \cdot d\theta$$

遇到奇点可以略一下看-看结果

老大是什么? 考虑

$$\begin{matrix} 1 & 2 & \dots & \infty \\ \pm 1 & \pm 2 & \dots & \pm \infty \\ 0 & \dots & \dots & \dots \end{matrix}$$

与谐振子类似.

会发光

$$= \frac{R}{8} \left\{ \frac{2\sinh[(\omega_1+1)t]}{\omega_1+1} + \frac{2\sinh[(\omega_1-1)t]}{\omega_1-1} + \frac{2\sinh[(\omega_2+1)t]}{\omega_2+1} + \frac{2\sinh[(\omega_2-1)t]}{\omega_2-1} \right\} \Big|_0^{2\pi}$$

$$\text{当 } (\omega_1 = m+n, \omega_2 = m-n)$$

而当出现 $\omega_1 - 1 = 0, \omega_2 - 1 = 0$ 的奇点后,

$$\langle x \rangle = \frac{R}{8} \cdot \pi, \text{ 存在发射与吸收.}$$

2-1-7: 实际上, 由于杂原子与畸变, 会偏离理想势阱, 产生发光. 将线性多烯看作有周期势垒 (原子) 的凹凸势箱会更好. (3')